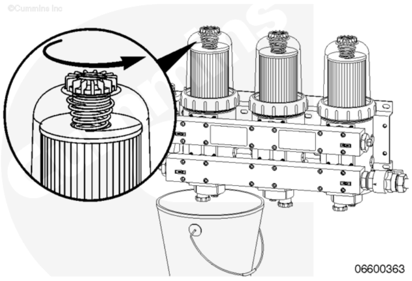
Trainings

Drain

with Electronically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



Industrial and Power Generation Applications

- The accumulated water **must** be drained daily.
- For fuel that has a high water content, more frequent draining will be necessary.
- Do **not** allow fuel to drain onto the ground. Drained fuel **must** be disposed of in accordance with local environmental regulations.

Shut the engine off.

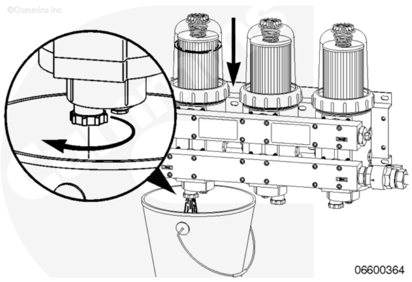
Place a suitable container under the fuel filter.

With the fuel supply valve closed, open the vent cap to break the air lock in the filter.

Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When fuel begins to flow out of the drain, close the drain valve.

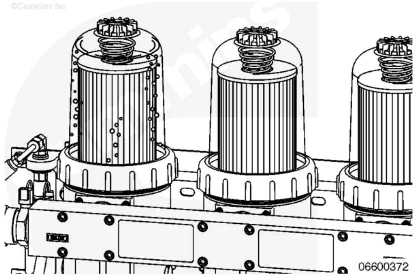
The drained liquids **must** be disposed of in accordance with local environmental regulations.

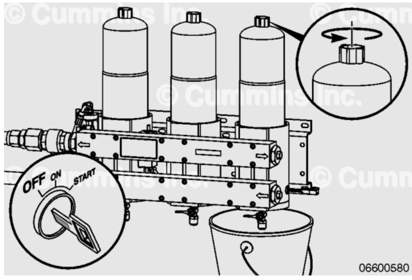


Close the vent cap.

Start the engine, raising the rpm for 1 minute to purge the air from the system.

If the engine fails to start, prime the filter to remove the excess air. Refer to Procedure 006-015 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-006-015-om.html)





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⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.

Note : Marine engines are equipped with two different Stage 1 filter systems. There are identified as Triplex or Quadplex (filters with individual shutoff valves) and Dual or Triple (filters without individual shutoff valves). Triplex and Quadplex filter applications allow the engine to remain in operation while the filters are changed.

Marine Dual/Triple Filter Application

- The accumulated water **must** be drained daily.
- For fuel that has a high water content, more frequent draining will be necessary.

Shut the engine off.

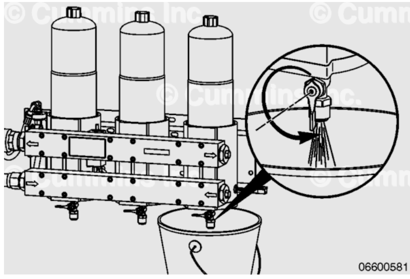
Place a suitable container under the fuel filter.

With the fuel supply valve closed, open the vent cap to break the air lock in the filter.

Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When the fuel begins to flow out of the drain, close the drain valve.

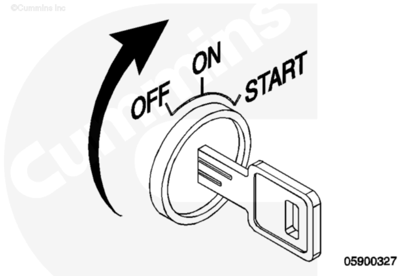
The drained liquids **must** be disposed of in accordance with local environmental regulations.



Close the vent cap.

Start the engine, raising the rpm for 1 minute to purge the air from the system.

If the engine does **not** start, perform the prime procedure to remove the excess air. Refer to Procedure 006-015 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-006-015-om.html)



⚠ WARNING ⚠

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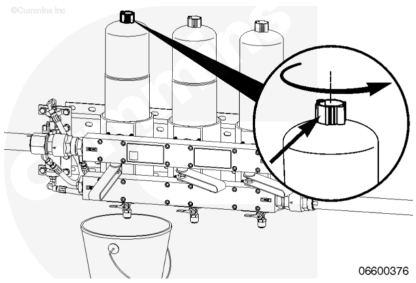


⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.

Marine Triplex/Quadplex Application

- The accumulated water **must** be drained daily.
- For fuel that has a high water content, more frequent draining will be necessary.

Move the filter selector valve handle to the OFF position for the filter to be drained.

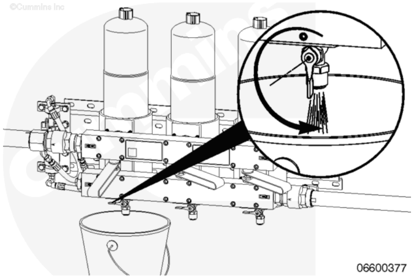
Place a suitable container under the fuel filter.

With the filter selector valve in the OFF position, open the vent cap to break the air lock in the filter.

Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When fuel begins to flow out of the drain, close the drain valve.

The drained liquids **must** be disposed of in accordance with local environmental regulations.

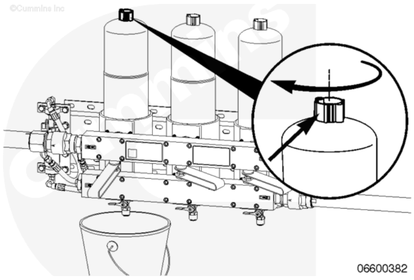


Close the vent cap.

If the engine was shut off, raise the rpm for 1 minute to purge the air from the system.

If the engine is in operation, slowly open the selector valve handle to the ON position, allowing extra time to purge air from the system.

If the engine fails to run properly, perform the prime procedure to remove the excess air. Refer to Procedure 006-015 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-006-015-om.html)



Maintenance Check

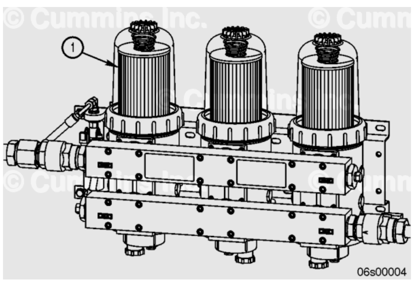
with Electronically Actuated Injector

Industrial and Power Generation Applications

Industrial and Power Generation application stage 1 filters contain a clear cover for inspection of the fuel level in the filter. Over time the fuel level in the filter will rise as the filter becomes contaminated.

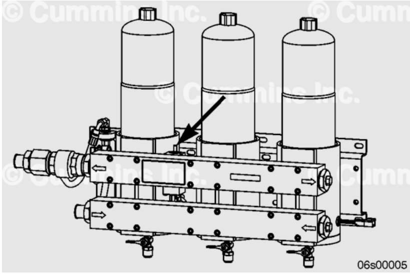
The filter **must** be replaced if the fuel level reaches the top of the filter element (1) prior to the standard maintenance interval. Check the fuel level in the filter. Replace if required. Refer to Procedure 006-075 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-006-075-om.html)

Note : The fuel level in the filter is not to be used as an extended life monitor. Filter life must not exceed the standard maintenance interval.



Marine Applications

Marine application engines are equipped with two different filter systems. They are identified as Non-Classed (1) (filters without individual shutoff valves) and Classed (2) (filters with individual shutoff valves).



The filter element in these filter systems uses a steel cover and therefore contains a pop-up indicator to indicate when the filters are contaminated before the standard maintenance interval. The pop up indicator is located on the right side of the filter housing on Non-Classed systems and between the two filters on Classed systems, as indicated by the arrows.

Note : The pop-up indicators are not shown on these graphics.

When the indicator pops up, all filters **must** be changed. Refer to Procedure 006-075 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-006-075-om.html)

Note : The pop up indicator is not to be used as an extended life monitor. Filter life must not exceed the standard maintenance interval.

Last Modified: 04-Dec-2013